

An aerial night view of a city, likely Shanghai, with a network overlay of glowing nodes and connecting lines. The city lights are visible, and the network lines are white and curved, connecting various points across the city. The sky is a deep blue, and the city lights are warm yellow and orange.

Wireless Networks

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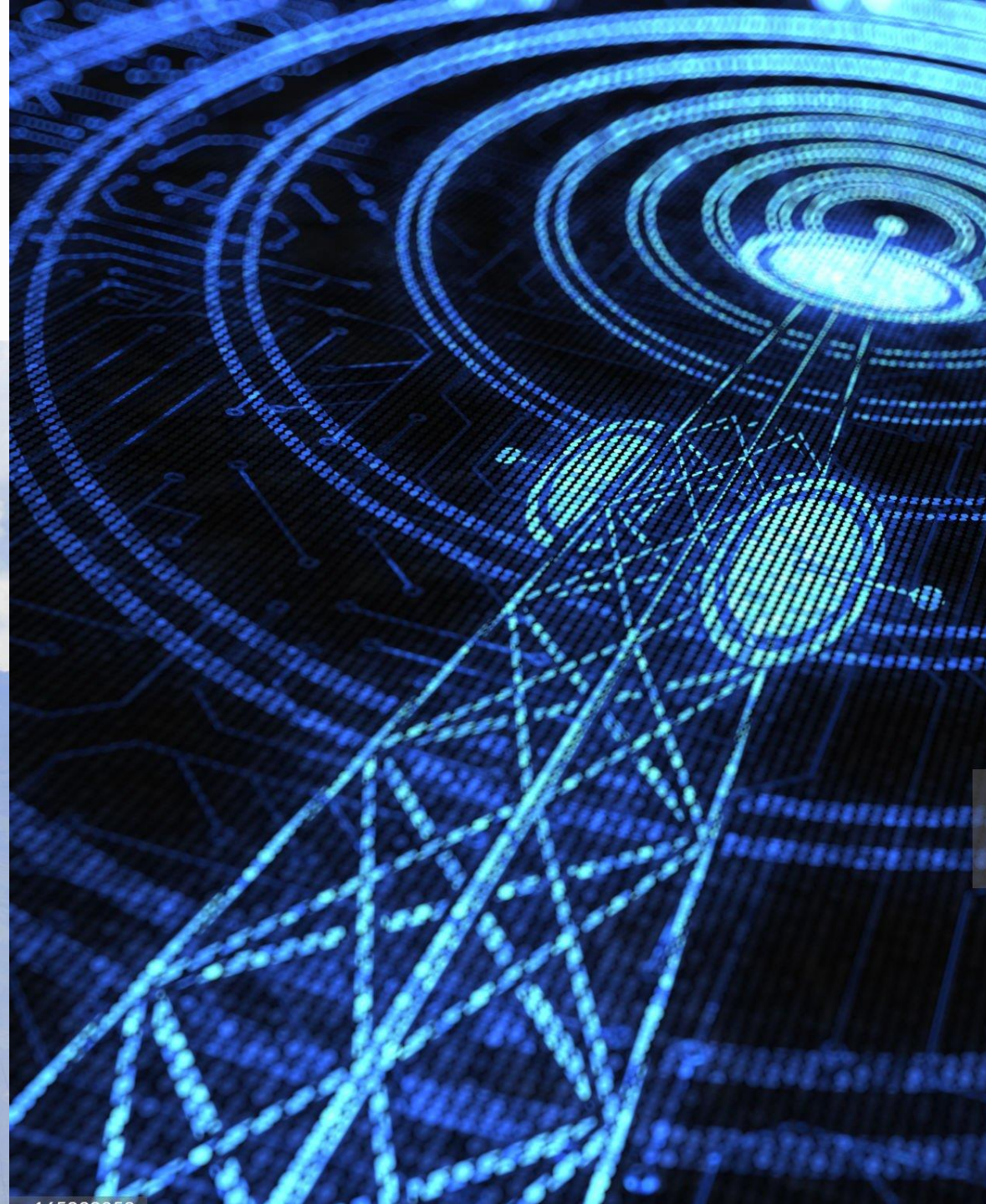
Suez Canal University

2024-2025

Wireless Networks

Lecture-4

Wireless Channel Model



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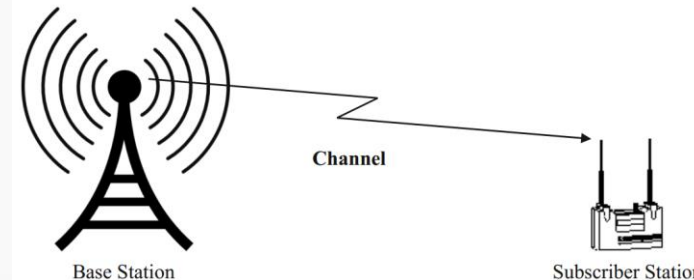
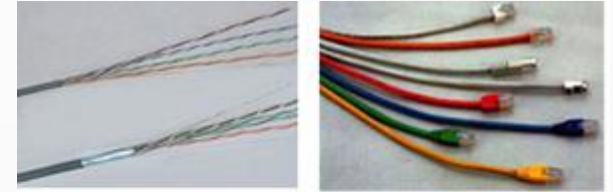
❖ **Multi-path problem**



Wireless Channel Model

What is 'channel' ?

- ☑ A **channel** is the physical medium through which a transmitted signal travels.
- **Standard for serial communication:** the RS232 cable (copper) acts as the channel.
- **Ethernet (Wired LAN):** the LAN cable serves as the channel between devices.
- **Mobile Communication:**
 - The air space and obstacles (e.g., mountains, buildings) form the channel.



Wireless Channel Model

What is 'channel' ?

☑ **Signal Propagation:**

- ✓ **Wired Communication:** Signals travel through a guided medium (e.g., copper wire, fiber-optic cable).
- ✓ **Wireless Communication:** Signals propagate through open mediums, encountering obstacles like walls, buildings, and trees.

☑ **Environmental Variability:**

- ✓ The wireless channel changes rapidly due to varying propagation conditions, and it is not fully controllable by system designers.

Wireless Channel Model

What is 'channel' ?

☑ **Wireless channel Impairments:**

- ✓ **Noise:** Unwanted random disturbances in the signal.
- ✓ **Path Loss:** Reduction in signal strength as it travels.
- ✓ **Shadowing:** Signal blockage due to obstacles.
- ✓ **Multipath Fading:** Signal distortion from reflections and multiple propagation paths.
- ✓ **Intersymbol Interference (ISI):** Overlapping of symbols due to delayed signals.
- ✓ **External Interference:** Disruptions from other wireless systems.
- ✓ **Source of Impairments:** These issues arise from analog processing at the transmitter/receiver and the propagation medium.

Wireless Channel Model

Followings are several common types of wireless channels.

Free Space, Fixed Transmitter/Receiver



Free Space, Moving Transmitter/Receiver



Reflecting wall, Fixed Transmitter/Receiver



Reflecting wall, Moving Transmitter/Receiver



Wireless Channel Model

Mathematical models of channels

- 1) Additive noise channel
- 2) Linear time-invariant filter channel
- 3) Linear time-variant filter channel

Wireless Channel Model

Additive noise channel

- ☑ **Noise** is an impairment in all communication systems, modelled as a **random process** with **fluctuations**.
- ☑ **Additive noise** refers to unwanted signals that get added to the original signal during transmission or processing.
- ☑ In 1948, Shannon showed that noise limits the number of bits that can be reliably transmitted.
- ☑ Noise is **superimposed on the desired signal**, obscuring or masking it.

Wireless Channel Model

Additive noise channel

☑ Types of Additive Noise:

2) Thermal Noise (Johnson-Nyquist Noise)

- ✓ Caused by the **random motion of electrons** in resistors and conductors.
- ✓ Present in all electronic devices and communication channels.
- ✓ Has a Gaussian distribution and is characterized as **Additive White Gaussian Noise (AWGN)**.

3) Shot Noise

- ✓ Occurs due to the **discrete nature** of electric charge.
- ✓ Common in semiconductor devices like diodes and transistors.
- ✓ More significant at low current levels.

Wireless Channel Model

Additive noise channel

☑ Types of Additive Noise:

4) Flicker Noise (1/f Noise)

- ✓ Has a power spectrum that decreases with frequency (higher at low frequencies).
- ✓ Found in semiconductor devices, vacuum tubes, and resistors.

5) Quantization Noise

- ✓ Introduced during analog-to-digital conversion (ADC).
- ✓ Arises due to the finite resolution of digital representation.

Wireless Channel Model

Additive noise channel

☑ Types of Additive Noise:

6) Intermodulation Noise

- ✓ Generated when multiple **signals mix nonlinearly**, producing unwanted frequency components.
- ✓ Common in radio and communication systems.

7) Impulse Noise

- ✓ Caused by sudden, short-duration disturbances (e.g., lightning, switching transients, or power line interference).

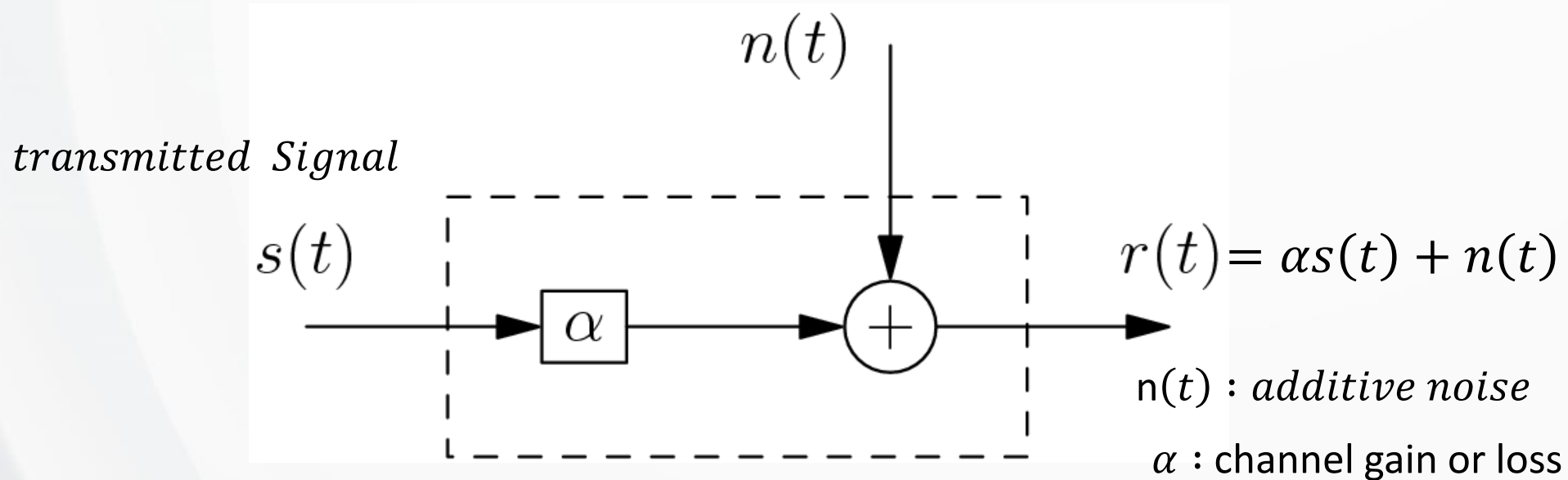
8) Phase Noise

- ✓ Related to **fluctuations** in the phase of a signal.

Wireless Channel Model

Additive noise channel

- ✓ The transmitted signal is subject to the **addition of random noise**.



- ✓ The noise $n(t)$ is typically modelled as a **random variable process**, with specific statistical properties (e.g., Gaussian, uniform, or Poisson).

Wireless Channel Model

Additive noise channel

☑ **AWGN (Additive White Gaussian Noise)**

- AWGN is a fundamental **noise model** used in communication systems to simulate the **effect of random noise** on transmitted signals.
- It represents the noise that is added to the signal.

☑ **Key Characteristics of AWGN:**

- **Additive**: The noise is simply added to a signal without modifying its characteristics.
- **White**: The noise power is uniformly distributed across all frequencies (i.e. **flat spectral density**)
- **Gaussian**: The noise amplitude follows a Gaussian (normal) distribution.
 - **Zero mean** ($\mu=0$): The noise has no systematic bias.
 - **Variance** (σ^2): The power of the noise is characterized by its variance.

Wireless Channel Model

Additive noise channel

Channel Capacity:

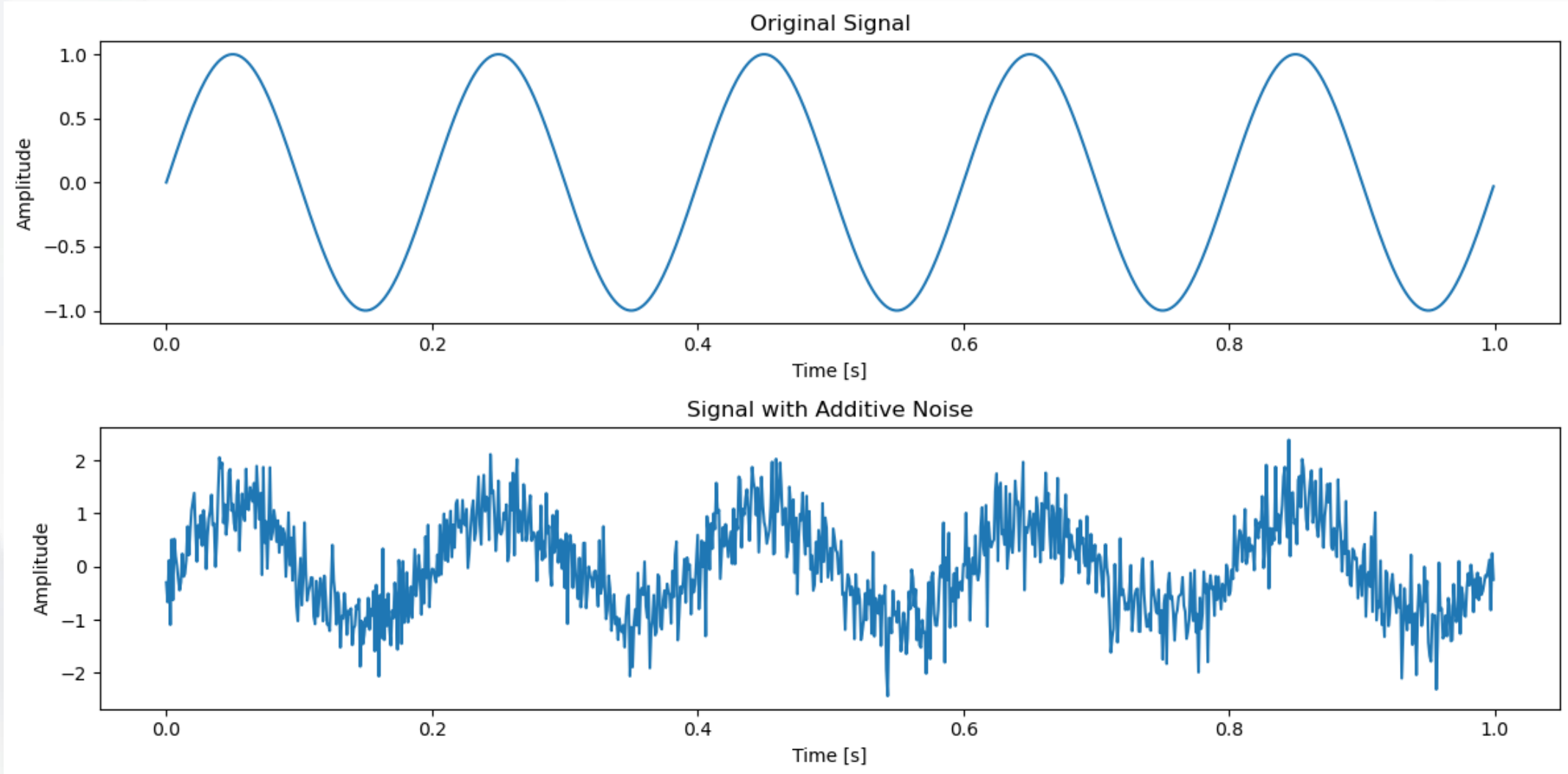
- For an AWGN channel, the channel capacity C , in bits per second (bps), is given by the Shannon theorem:

$$C = B \log_2 \left(1 + \frac{P}{N_0 B} \right)$$

- B : Bandwidth of the channel.
- P : Signal power.
- N_0 : Noise power spectral density.

Wireless Channel Model

Additive noise channel



Wireless Channel Model

Linear time-variant channel

☑ Challenges in Wireless Communications

1. Temporal Dispersion

1. Caused by **multipath propagation** in the wireless channel.
2. Leads to distortion of the transmitted signal.

2. Multipath Propagation

1. Occurs due to multiple paths between the **transmitter and receiver**.
2. Results from **reflection, penetration, diffraction, and scattering**.
3. Causes signals to arrive with different **delays, attenuations, and phase shifts**.
4. Leads to **dispersive distortion** of the transmitted signal.

Wireless Channel Model

Linear time-variant channel

- ☑ The relationship between input and output **remains linear**, which is reasonable to describe **far-field** propagation.
- ☑ Far-Field Propagation refers to the transmission of signals over **long distances**.
- ☑ Due to the motion of the transmitter, receiver, or objects in the environment, the channel model is time varying because sources of **multipath** can change over time because of.

Wireless Channel Model

Linear time-variant channel

- ☑ Given the linear time-varying assumption, and in the **absence of noise**, the received continuous-time signal $y(t)$ is related to the transmitted continuous-time signal $x(t)$ through a superposition integral between the time-varying impulse response $h_c(t, \tau)$ via

$$y(t) = \int_{-\infty}^{\infty} h_c(t, \tau) x(\tau) d\tau.$$

- ☑ The time-varying impulse response $h_c(t, \tau)$ captures all the effects of multipath and filtering in the transmission path as well as the variation of the transmission path over time.

Wireless Channel Model

Linear time-variant channel

- ☑ Given the linear time-varying assumption, and in the **presence of noise**, the received continuous-time signal $y(t)$ is related to the transmitted continuous-time signal $x(t)$ through a superposition integral between the time-varying impulse response $h_c(t, \tau)$ via

$$y(t) = \int_{\tau=-\infty}^{-\infty} x(t) * h(t - \tau) d\tau + n(t)$$

- Where $s(t) * h(t)$: The convolution of the channel's impulse response $h(t)$ with the transmitted signal $s(t)$.
- $n(t)$: Additive noise.

Wireless Channel Model

Linear time-invariant channel

☑ Neglecting Channel Time Variation in Signal Processing

- Most wireless systems assume **channel time variation is negligible** over short intervals.
- This simplifies **signal processing algorithm design**.

☑ Challenges of Time-Varying Channels

- Time-varying channels $h_c(t, \tau)$ are difficult to **estimate, track, and equalize**.
- To manage this, information is sent in **short bursts**, shorter than the **coherence time** (rate of channel variation).

Wireless Channel Model

Linear time-invariant channel

☑ Time-Invariance Assumption for Modelling

- Wireless channels are modelled as **linear time-invariant (LTI) systems**.
- This assumption holds when the **channel remains constant long enough** for estimation and compensation.

☑ Mathematical Representation of LTI Systems

- The **impulse response** $h_c(t)$ characterizes the LTI channel.
- Input and output are related through a **convolution integral** in continuous time.

$$y(t) = \int_{-\infty}^{\infty} h_c(\tau)x(t - \tau)d\tau$$

Wireless Channel Model

Linear time-invariant channel

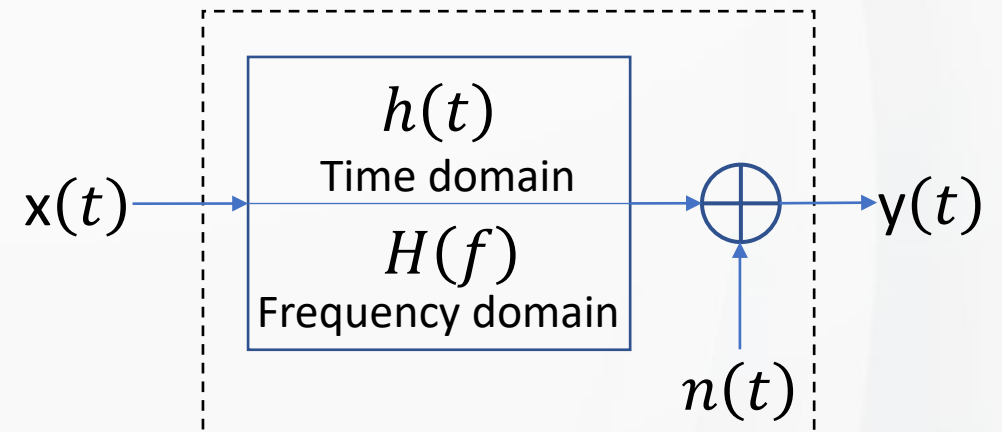
✓ In **discrete time**, the convolution integral becomes a **summation**:

- Given input $x[n]$, output $y[n]$, and impulse response $h[n]$, the system follows convolution.

$$y[n] = \sum_{\ell=-\infty}^{\infty} h[\ell]x[n - \ell]$$

Channel

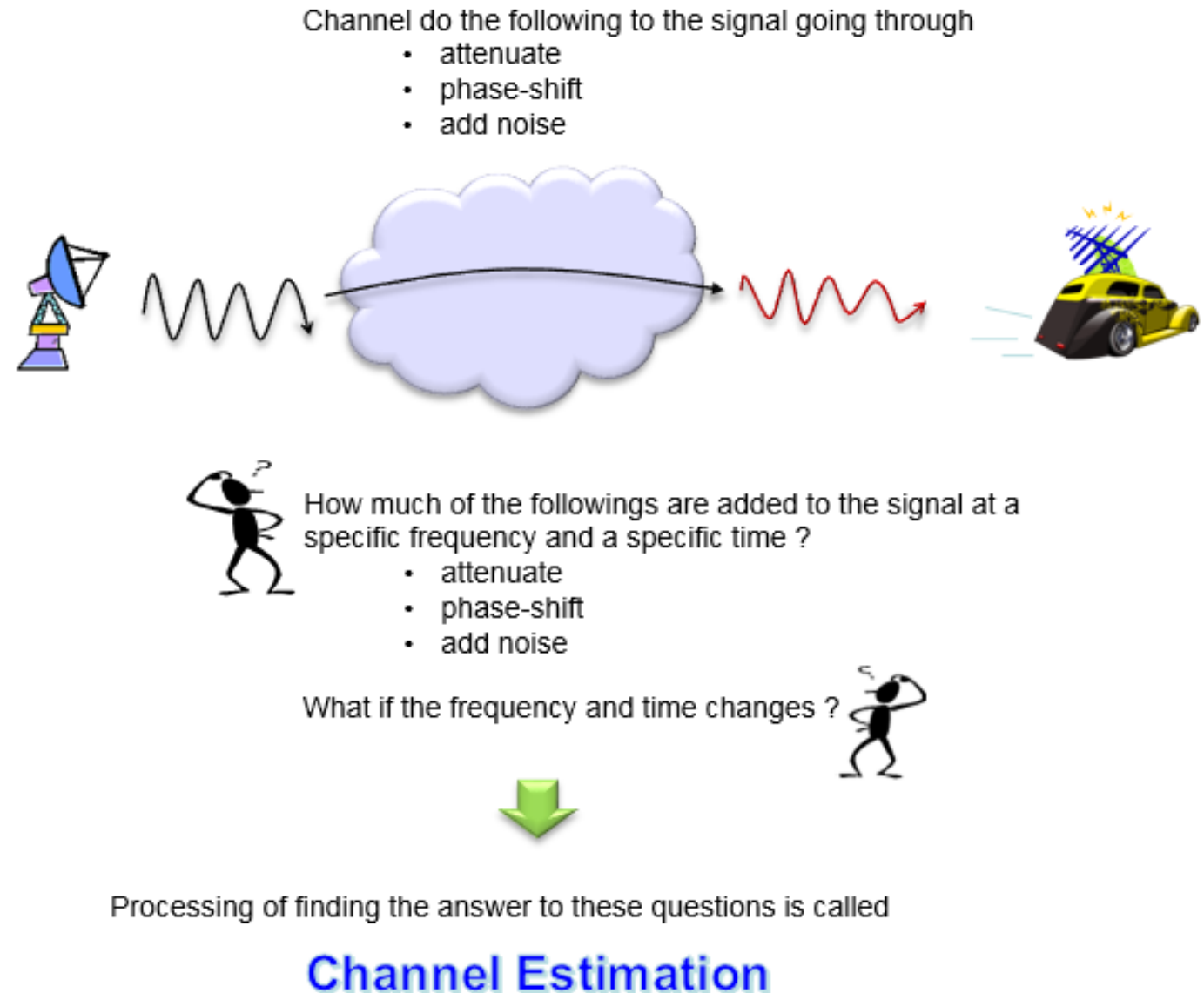
$$\begin{aligned} h(t) &\xrightarrow{\text{FTT}} H(f) \\ h(t) &\xleftarrow{\text{IFTT}} H(f) \end{aligned}$$



Wireless Channel Model

Channel Estimation

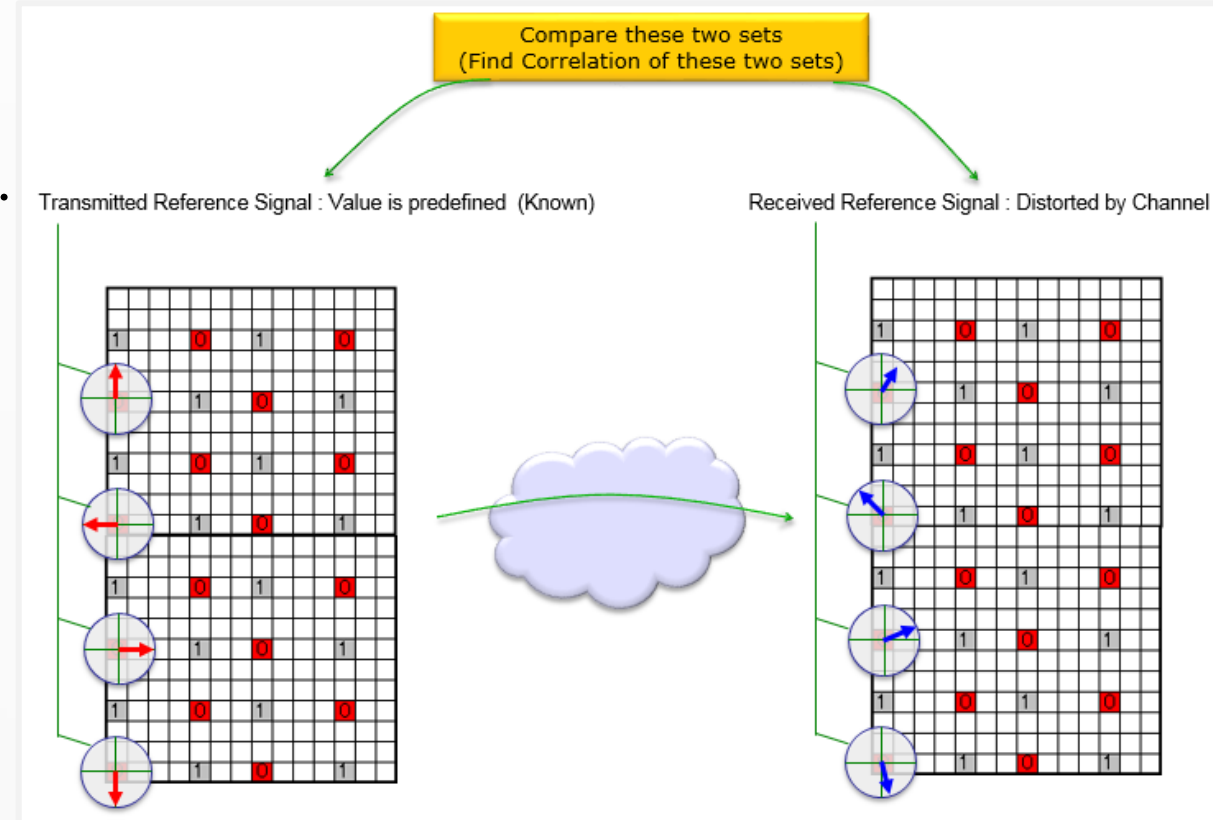
We **need** to find a way to **predict** or **estimate** the signal that may be received at a given location under certain transmissions.



Wireless Channel Model

Channel Estimation: General Algorithm

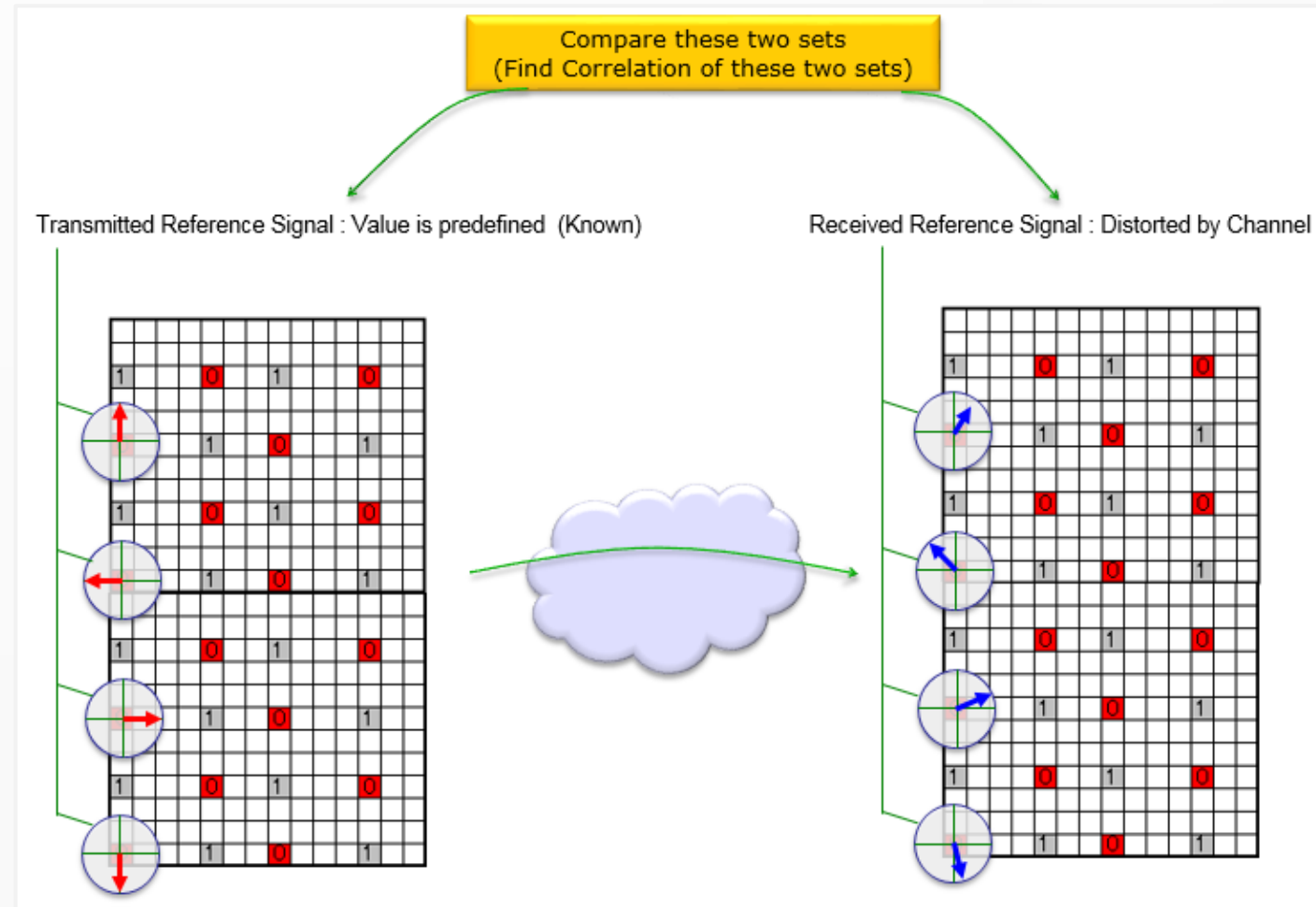
- 1) A predefined **pilot signal** or **reference signal** is inserted into the transmission.
- 2) The receiver already knows this reference signal.
- 3) As the signal propagates through the channel, it experiences:
 - Attenuation (signal weakening)
 - Phase shifts
 - Noise addition (thermal noise, interference)
 - Multipath fading (signals arriving at different times due to reflections)



Wireless Channel Model

Channel Estimation: General Algorithm

- 1) The receiver captures the distorted reference signal.
- 2) It detects and decodes the received pilot signal.
- 3) The receiver compares the received reference signal with the originally transmitted reference signal.
- 4) This helps identify how the channel modified the signal.



Free space Propagation Model

■ Path Loss

- ☑ Signal loses power during travel.
- ☑ This is called path loss or signal attenuation.
- ☑ Received power is a **fraction** of the original transmitted power.
- ☑ Path loss **increases** with **distance** between transmitter and receiver.
- ☑ Path loss **must be estimated** for **designing** wireless links.
- ☑ Different path loss models are used for different scenarios.
 - **Frii's model**: for free space without reflections.
 - **2-ray model**: considers ground reflections; more practical.

Free space Propagation Model

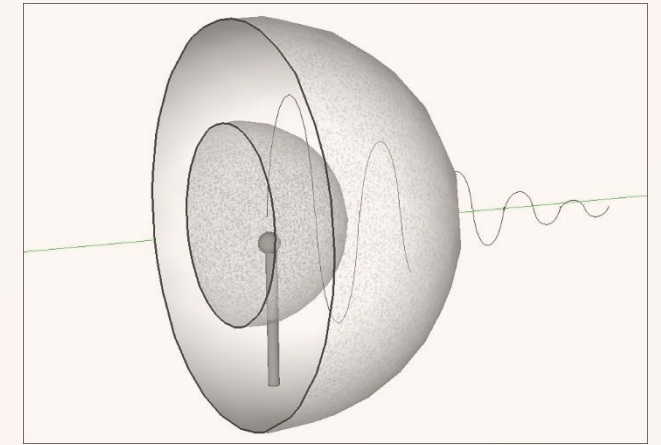
■ Frii's law

The **free space path loss** is the **loss** in signal strength as it travels through **free space**.

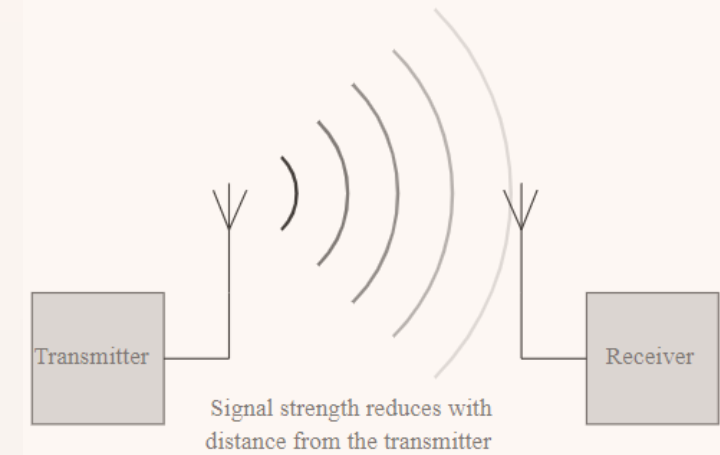
- In isotropic antenna, the RF energy travels in **every direction** over a sphere in free space.
- That energy gets **weaker** as the distance from the antenna increases.

$$\text{Signal level} = \frac{k}{d^2}$$

- Where:
 - **k** is constant
 - **d** is the distance from the transmitter.



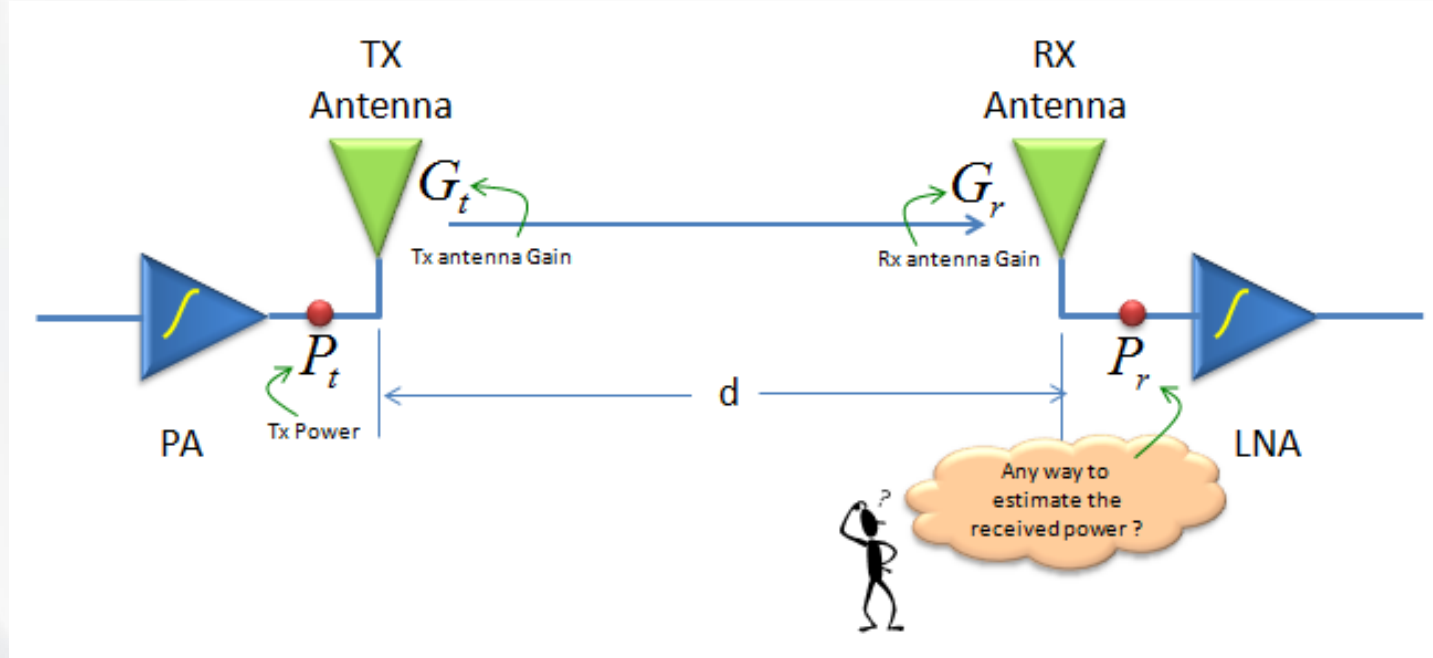
Free Space Loss Due to Wave Spreading



Free space Propagation Model

■ Frii's law

- ✓ In free space without any **absorbing** or **reflecting** objects, the path loss depends on the **distance** as well as on the frequency (or wavelength) according to the following Frii's law:



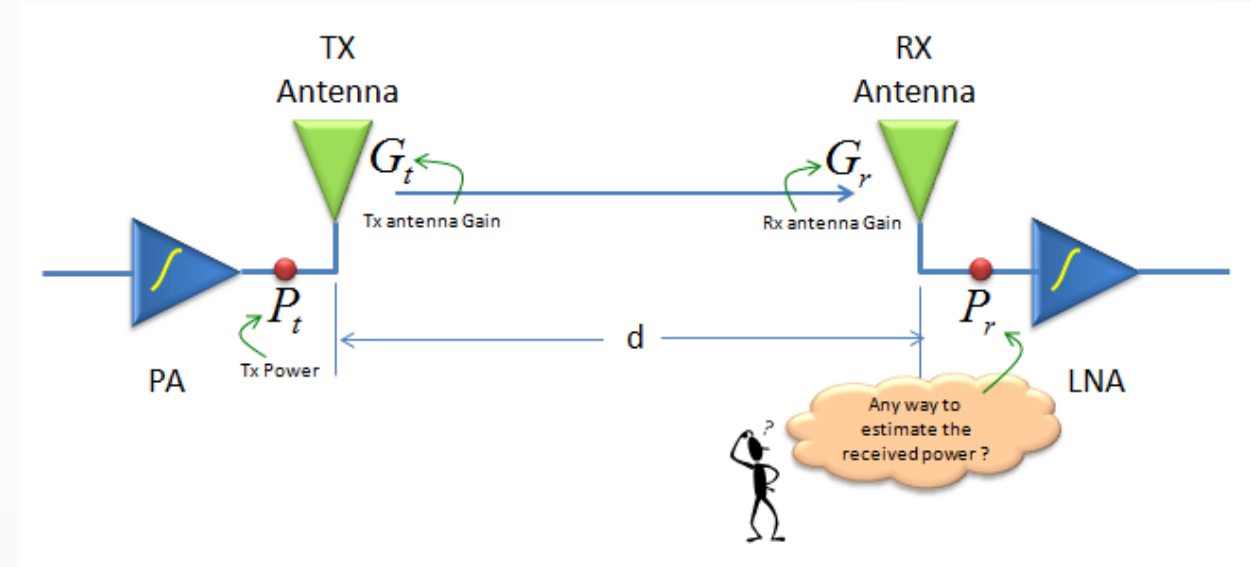
$$P_R = P_T G_T G_R \left(\frac{\lambda}{4\pi d} \right)^2 = P_T G_T G_R \left(\frac{c}{4\pi f d} \right)^2$$

- ✓ Where P_R and P_T are the received and transmitted powers (in Watts)
- ✓ G_T and G_R are transmitter and receiver antenna gains in linear scale,

Free space Propagation Model

■ Frii's law

$$P_R^{dB} = P_T^{dB} + G_T^{dB} + G_R^{dB} + 10\log_{10}\left(\frac{\lambda}{4\pi d}\right)^2$$



- The path loss measured in dB as:

$$Path Loss = P_T^{dB} - P_R^{dB} = -G_T^{dB} - G_R^{dB} - 10\log_{10}\left(\frac{\lambda}{4\pi d}\right)^2$$

Free space Propagation Model

■ Frii's law

For isotropic antennas (G_T^{dB} and G_R^{dB} are both 0 dB), path loss is reduced to the following simple formula:

$$\begin{aligned}\text{Path loss}(dB) &= -10\log_{10}\left(\frac{\lambda}{4\pi d}\right)^2 = 20\log_{10}\left(\frac{4\pi d}{\lambda}\right) = 20\log_{10}\left(\frac{4\pi fd}{c}\right) \\ &= 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right) \\ &= 20\log_{10}(d) + 20\log_{10}(f) - 147.55\end{aligned}$$

Free space Propagation Model

■ Frii's law

Example 2

If 50 W power is applied to a 900 MHz frequency at a transmitter, find the receive power at a distance of 100 meter from the transmitter (assume free space path loss with unit antenna gains).

Solution

Unit antenna gain means: $G_T = G_R = 1$.

We have $d = 100$ m, $f = 900 \times 10^6$ Hz, $P_T = 50$ W, $c = 3 \times 10^8$ m/sec, and $\pi = 3.14$

$$P_R = P_T \left(\frac{c}{4\pi fd} \right)^2 = 3.5 \mu W$$

Free space Propagation Model

■ Frii's law

Example 3

What is the received power in dBm at 10 meters from a 2.4 GHz Wi-Fi router transmitting with 100 mW of power (assume free space path loss with unit antenna gains)?

Solution

Unit antenna gain means: $G_T = G_R = 0$ dBm.

We have $d = 10$ m, $f = 2.4 \times 10^9$ Hz, $P_T = 100$ mW = 20 dBm, $c = 3 \times 10^8$ m/sec, and $\pi = 3.14$

$$Pathloss = 20 \log_{10} \left(\frac{4\pi fd}{c} \right) dB = 60 dB$$

$$P_R = P_T - pathloss = 20 - 60 = -40 \text{ dBm}$$

Free space Propagation Model

■ Frii's law

Another Solution

$$\begin{aligned}\text{Path loss}(dB) &= -10\log_{10}\left(\frac{\lambda}{4\pi d}\right)^2 = 20\log_{10}\left(\frac{4\pi d}{\lambda}\right) = 20\log_{10}\left(\frac{4\pi fd}{c}\right) \\ &= 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right) \\ &= 20\log_{10}(d) + 20\log_{10}(f) - 147.55\end{aligned}$$

Unit antenna gain means: $G_T = G_R = 0$ dBm.

We have $d = 10$ m, $f = 2.4 \times 10^9$ Hz, $P_T = 100$ mW = 20 dBm, $c = 3 \times 10^8$ m/sec, and $\pi = 3.14$

$$\text{Path Loss in DB} = 20\log_{10}(d) + 20\log_{10}(f) - 147.55$$

$$\text{Path Loss in DB} = 20\log_{10}(10) + 20\log_{10}(2400000000) - 147.55$$

$$\text{Path Loss in DB} = 20 + 187.604 - 147.55$$

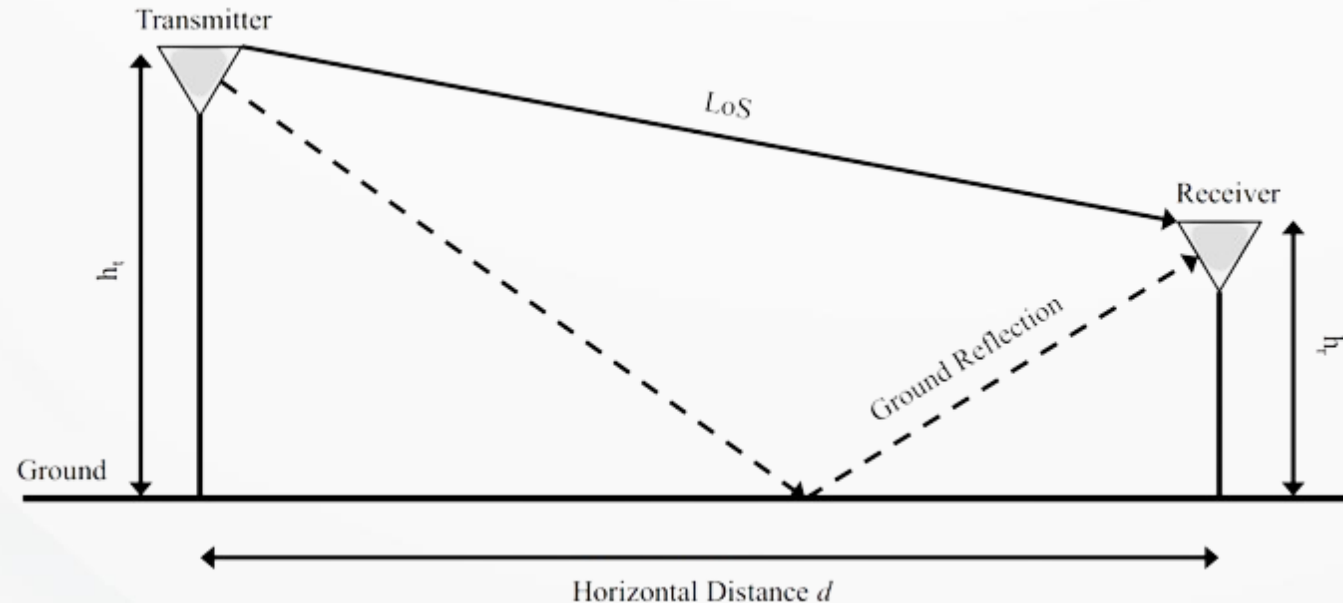
$$\text{Path Loss in DB} = 20 + 187.604 - 147.55 \approx 60 \text{ dB}$$

$$P_R = P_T - \text{Path Loss} = 20 - 60 = -40 \text{ dBm}$$

Free space Propagation Model

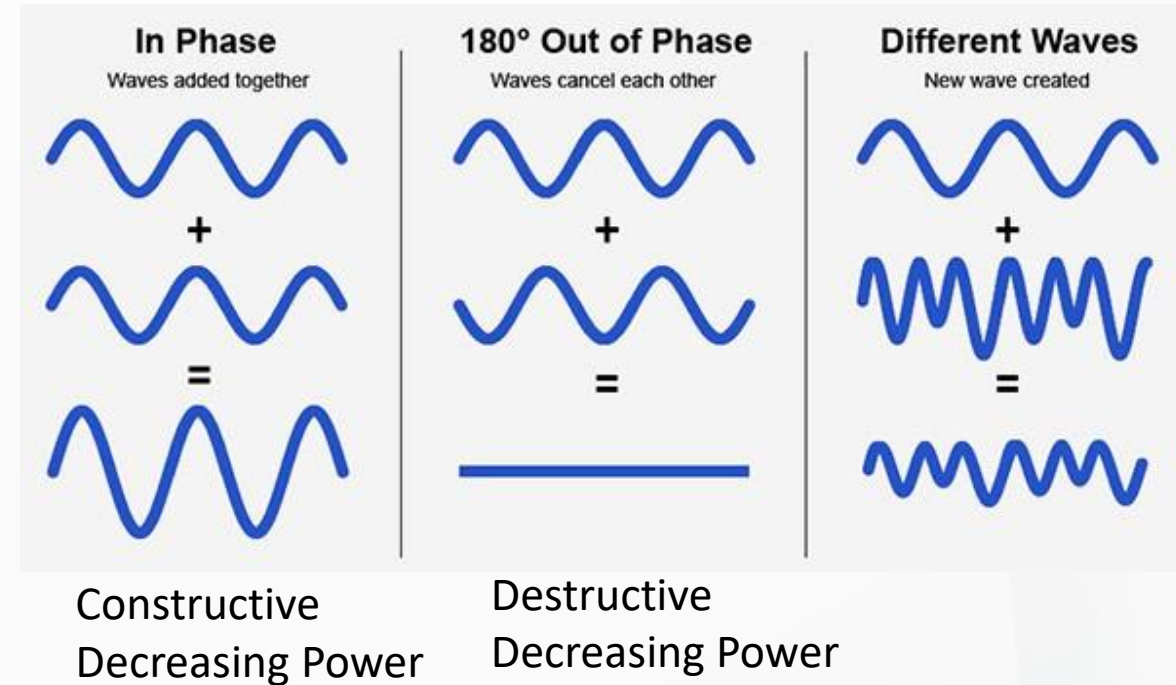
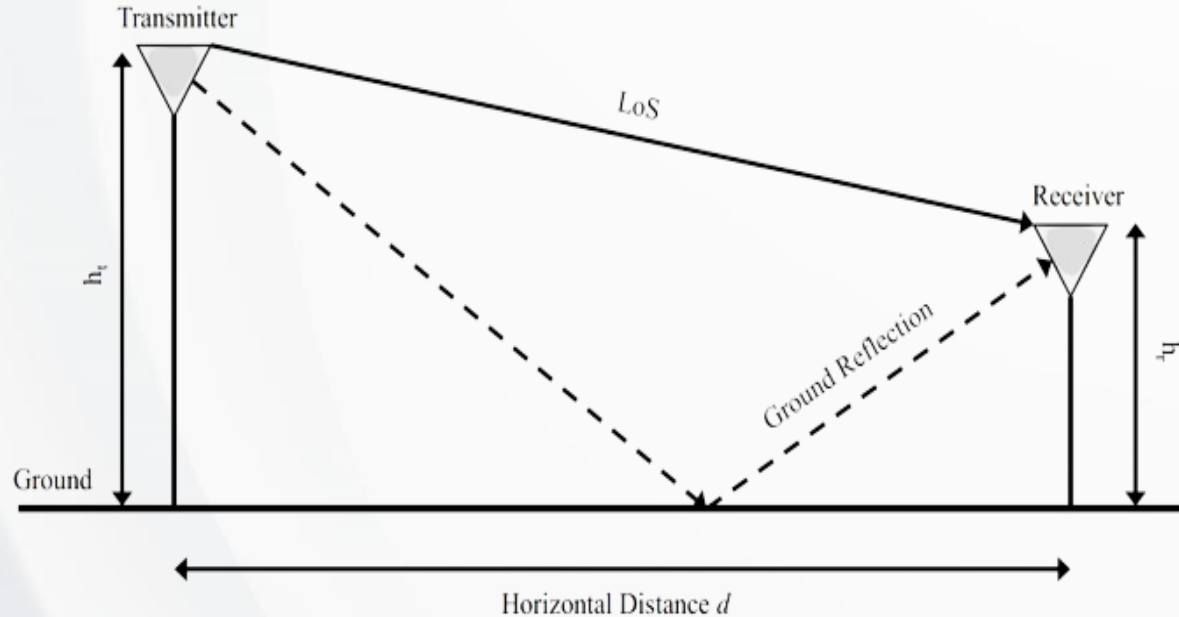
■ 2-ray Model (Ground reflection model)

- The 2-Ray Propagation Model is a radio wave propagation model that takes into account two primary paths for signal transmission between a transmitter and a receiver:
 - ✓ **Direct line-of-sight (LOS) path.**
 - ✓ **Ground-reflected path.**
- Accurate than the simpler **free-space model**, which only accounts for the LOS path.



Free space Propagation Model

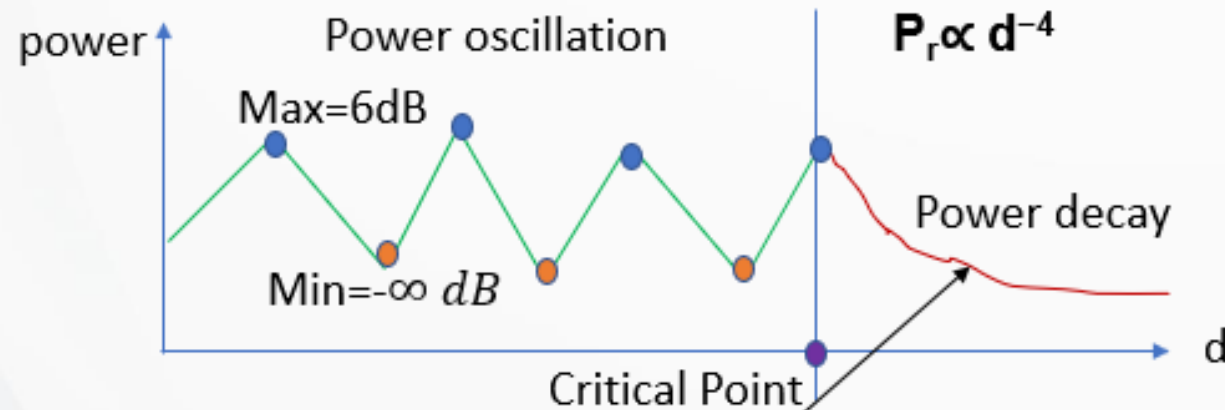
■ 2-ray Model (Ground reflection model)



Free space Propagation Model

■ 2-ray Model (Ground reflection model)

- Describes the rapid decrease in signal power with increasing distance in the 2-ray propagation model under far-field conditions.



$$PL(dB) = 40\log_{10}(d) - 10\log_{10}(G_t) - 10\log_{10}(G_r) - 20\log_{10}(h_t) - 20\log_{10}(h_r)$$

Free space Propagation Model

■ 2-ray Model (Ground reflection model)

Example: A **2m** tall user is holding his smartphone at half of his height while standing **500m** from a **10m**-high base station. The base station is transmitting a **1.8 GHz** signal using a transmission power of **30 dBm**. What is the received power (in dBm) at the smartphone? Assume unit gain antennas.

Sol.: unit gain antennas has 0 dB.

We have $h_t = 10$ m, $h_r = 1$ m, $d = 500$ m, $f = 1.8 \times 10^9$ Hz, $P_t = 30$ dBm

$$PL(dB) = 40 \log_{10}(d) - 10 \log_{10}(G_t) - 10 \log_{10}(G_r) - 20 \log_{10}(h_t) - 20 \log_{10}(h_r)$$

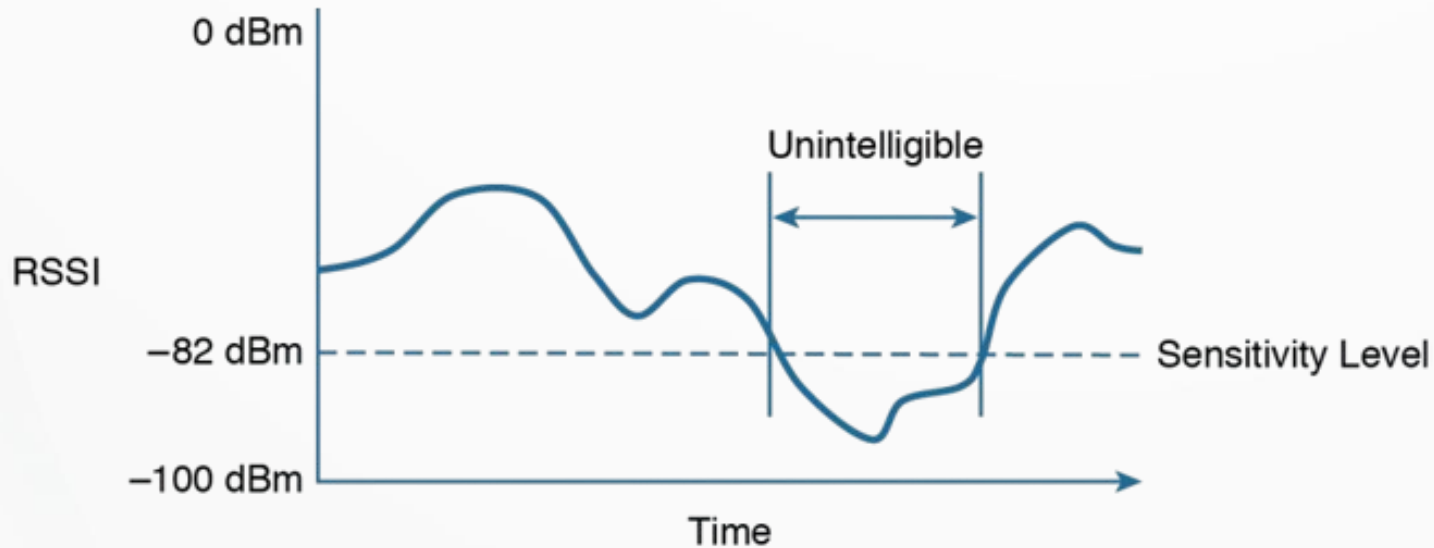
$$PL(dB) = 40 \log_{10}(500) - 0 - 0 - 20 \log_{10}(10) - 20 \log_{10}(1)$$

$$PL(dB) = 40 \times 2.7 - 0 - 0 - 20 \times 1 - 20 \times 0 = 108 - 20 = 88$$

$$\text{Received power} = 30 - 88 = -58 \text{ dBm (approximated)}$$

Receiver Sensitivity

- ❖ Receiver sensitivity refers to the minimum **received signal strength indicator** (RSSI) required for that receiver to be able to decode the received signal.



- ❖ **Noise**, **bandwidth**, and **modulation** affect the receiver sensitivity.
- ❖ For example, **Bluetooth** specifies that: **RSSI must be of -70 dBm to -82 dBm**

Receiver Sensitivity

Example 4

To increase the coverage with low transmit power, a manufacturer produced Bluetooth chipsets with a receiver sensitivity of -80 dBm. What is the maximum communication range that can be achieved for this chipset for a transmit power of 1 mW? Assume Free Space Path Loss with unit antenna gains.

Solution

Bluetooth frequency $f = 2.4$ GHz, $P_T = 1$ mW, $P_R = -80$ dBm $= 10^{-8}$ mW

We have

$$P_R = P_T \left(\frac{c}{4\pi df} \right)^2$$

$$d = \frac{c}{4\pi f} \sqrt{P_T / P_R}$$

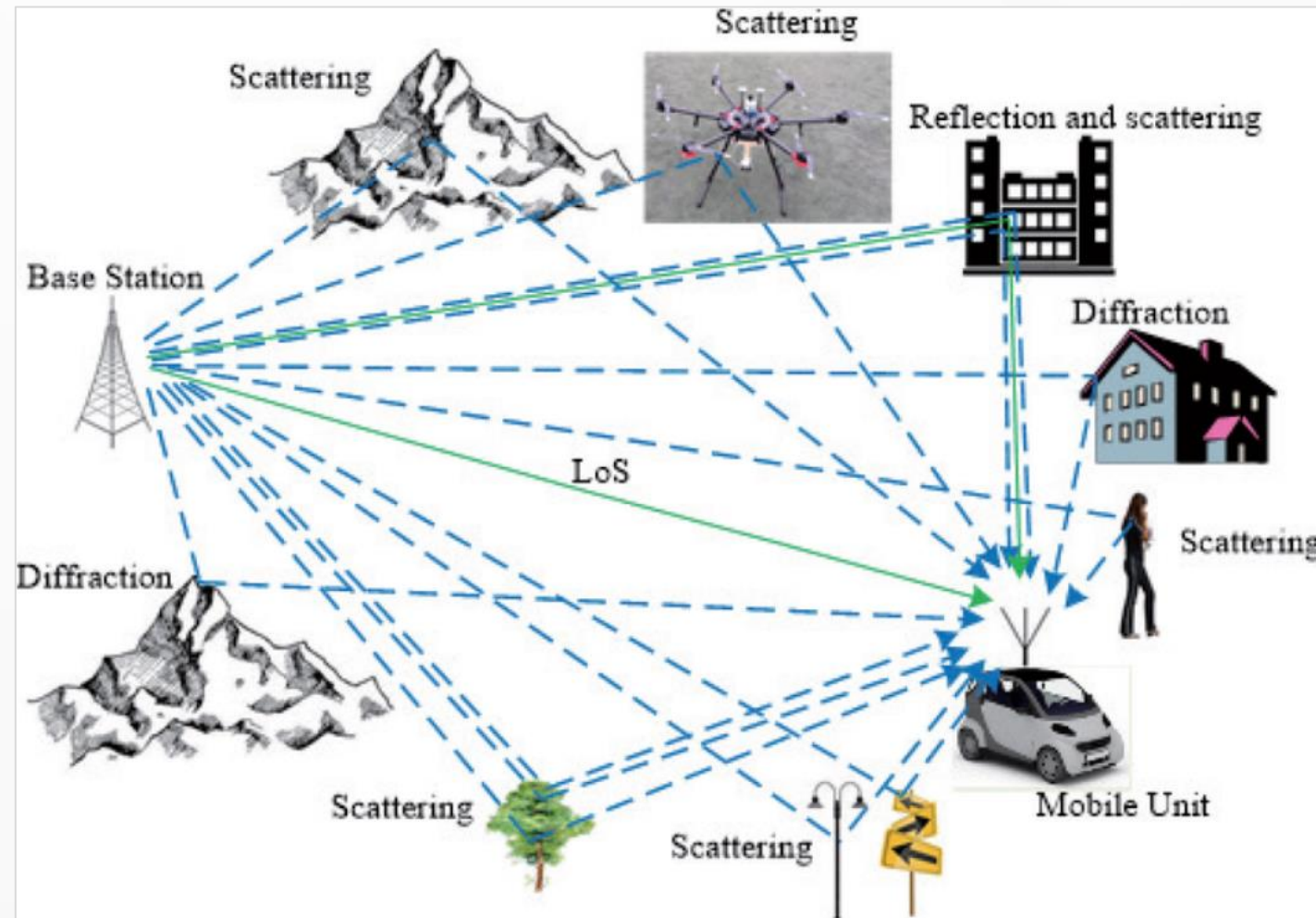
$$= 99.5 \text{ meter}$$

Multipath Propagation Problem

✓ The wireless multipath problem is a significant challenge in wireless communication systems.

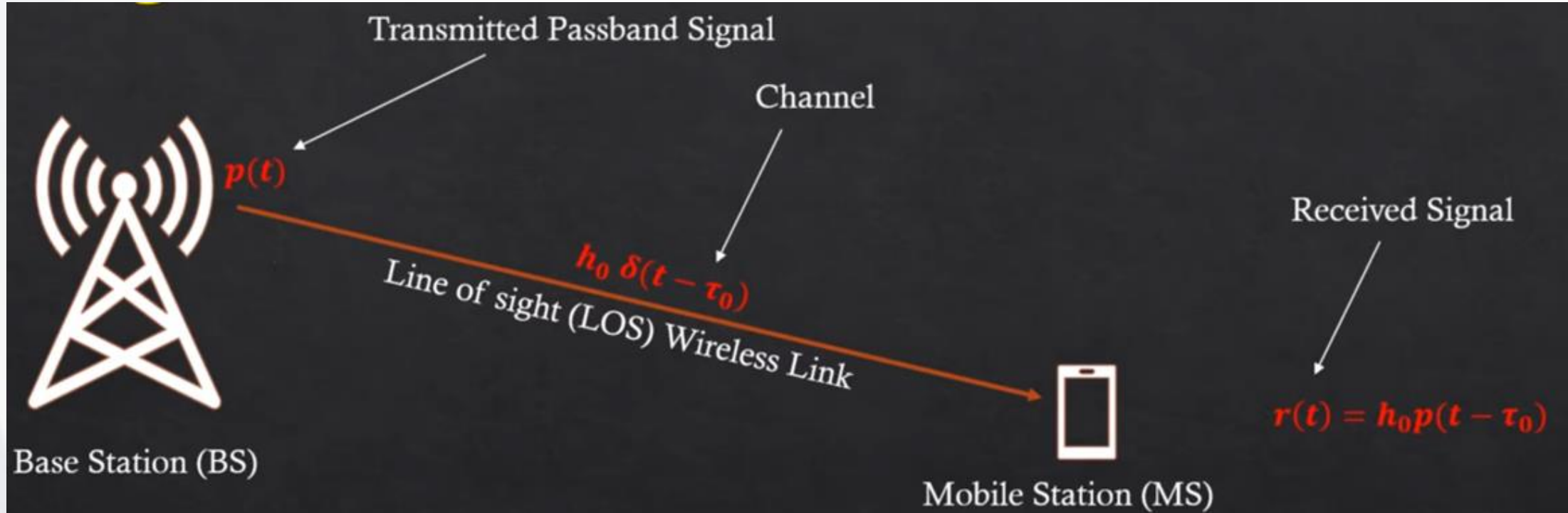
It occurs when a transmitted **signal** travels from the **transmitter** to the **receiver** via multiple **paths** due to **reflection**, **diffraction**, and **scattering** caused by obstacles such as buildings, trees, and other objects in the environment.

✓ These multiple paths result in multiple copies of the signal arriving at the receiver at different times, with different amplitudes, and phases.



Multipath Propagation Problem

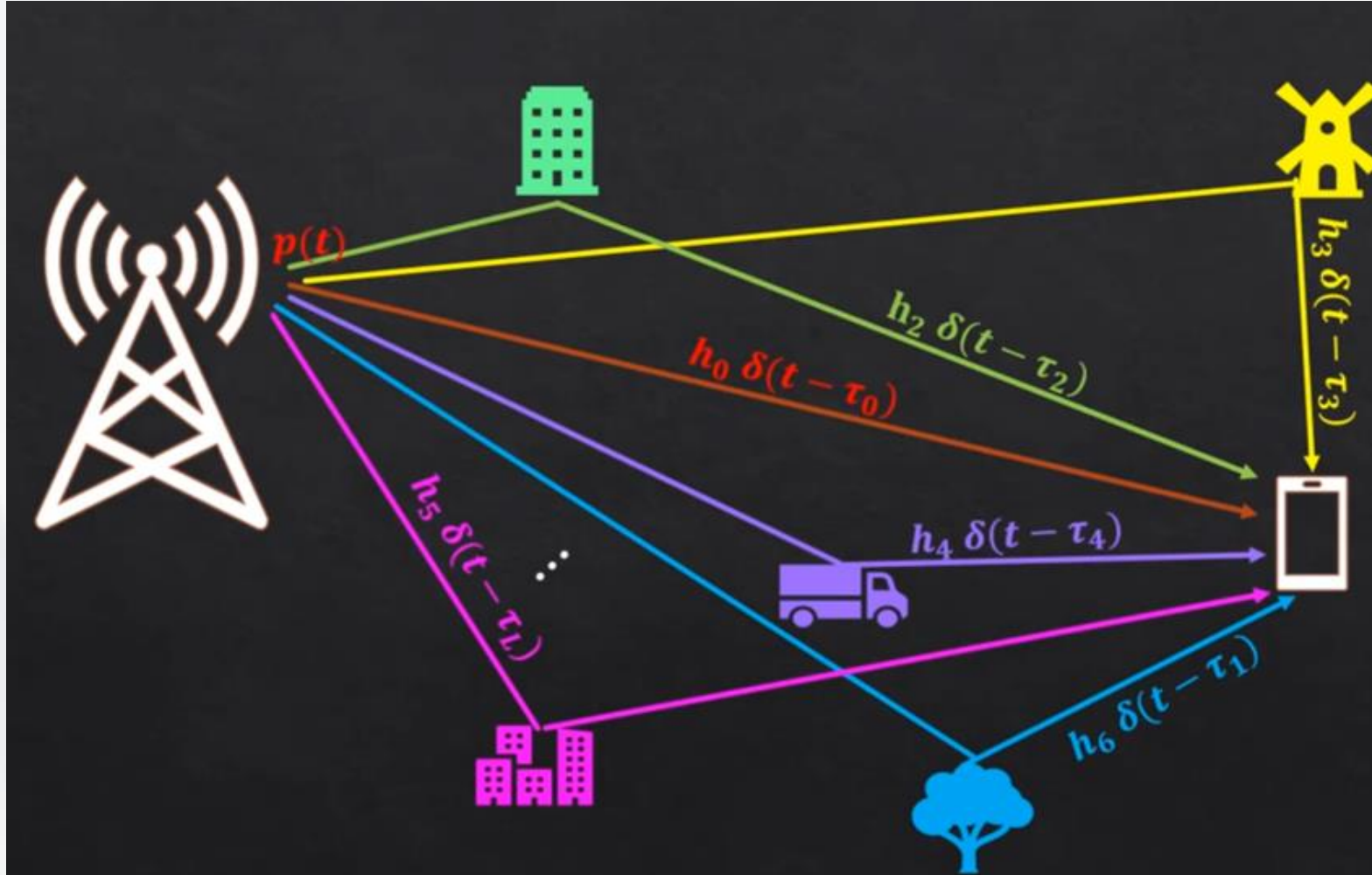
Single path channel Model



- ❖ Received signal is just attenuated by a factor h_0 and delayed by some time τ_0
 - ✓ factor h_0 depends on propagation medium, transmitted frequency, separation between BS and MS, Tx/Rx gains etc.
 - ✓ Delay, τ_0 , primarily depend on speed of EM wave in propagation medium (close to speed of light), separation between BS and MS.

Multipath Propagation Problem

Multi path channel Model



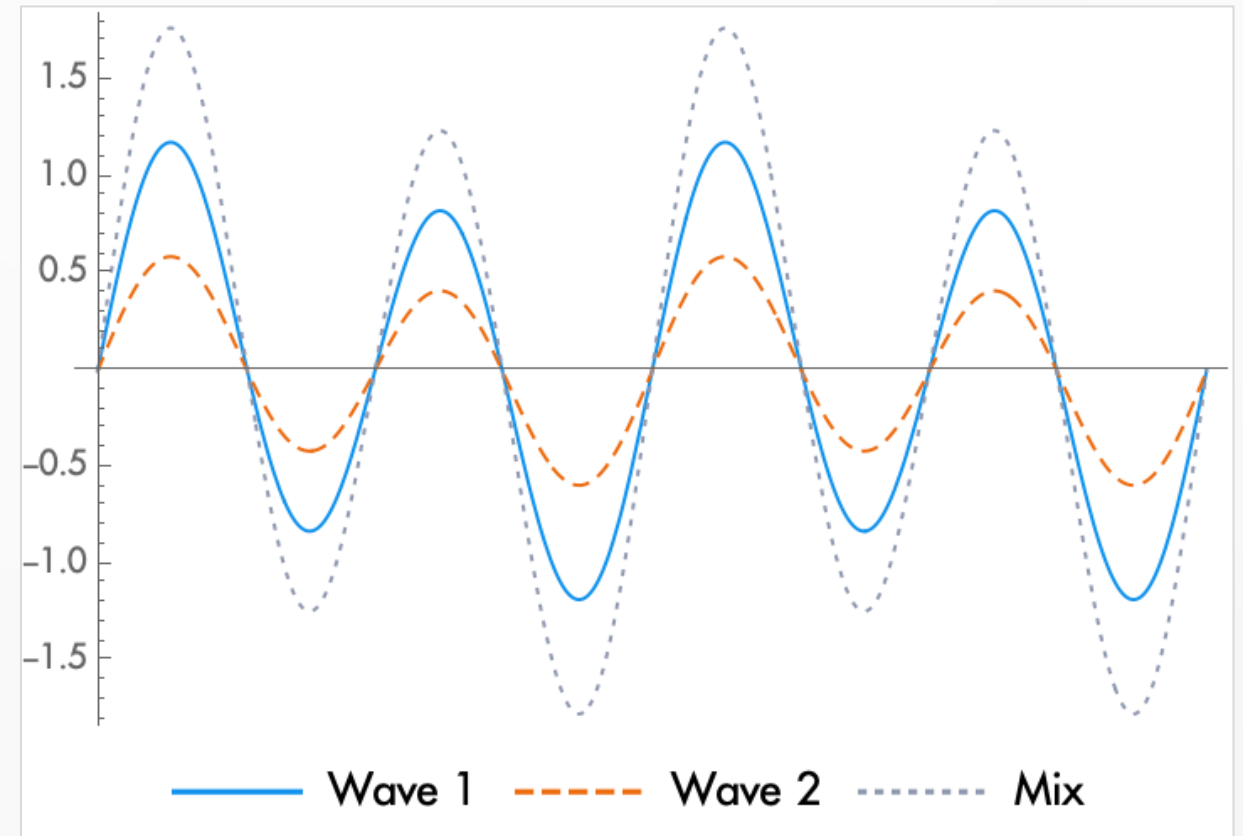
- ❖ In practice there could be several reflected/scattered paths reaching to MS from various objects
- ❖ Each path can have different attenuation and delays.

$$r(t) = \sum_{l=0}^L h_l p(t - \tau_l)$$

- ❖ Received signal is composed of superposition of multiple paths

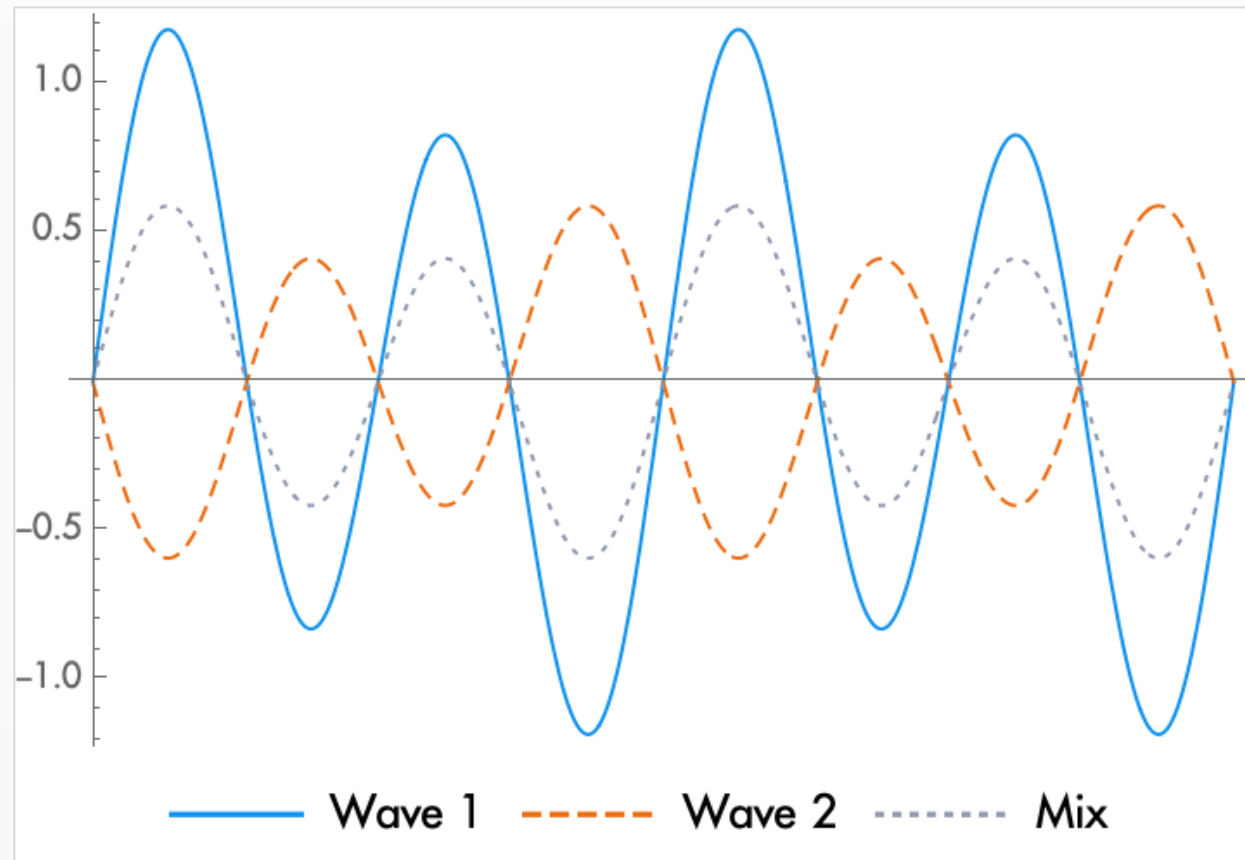
Multipath Propagation Problem

- ☑ The received signal has large time variation (dispersion) due to adding amplitude copies of the same signal.
- ☑ Signals can have different phases, frequencies, and shapes.
- ☑ When peaks of waves align, they enhance each other, increasing amplitude (**constructive interference**).



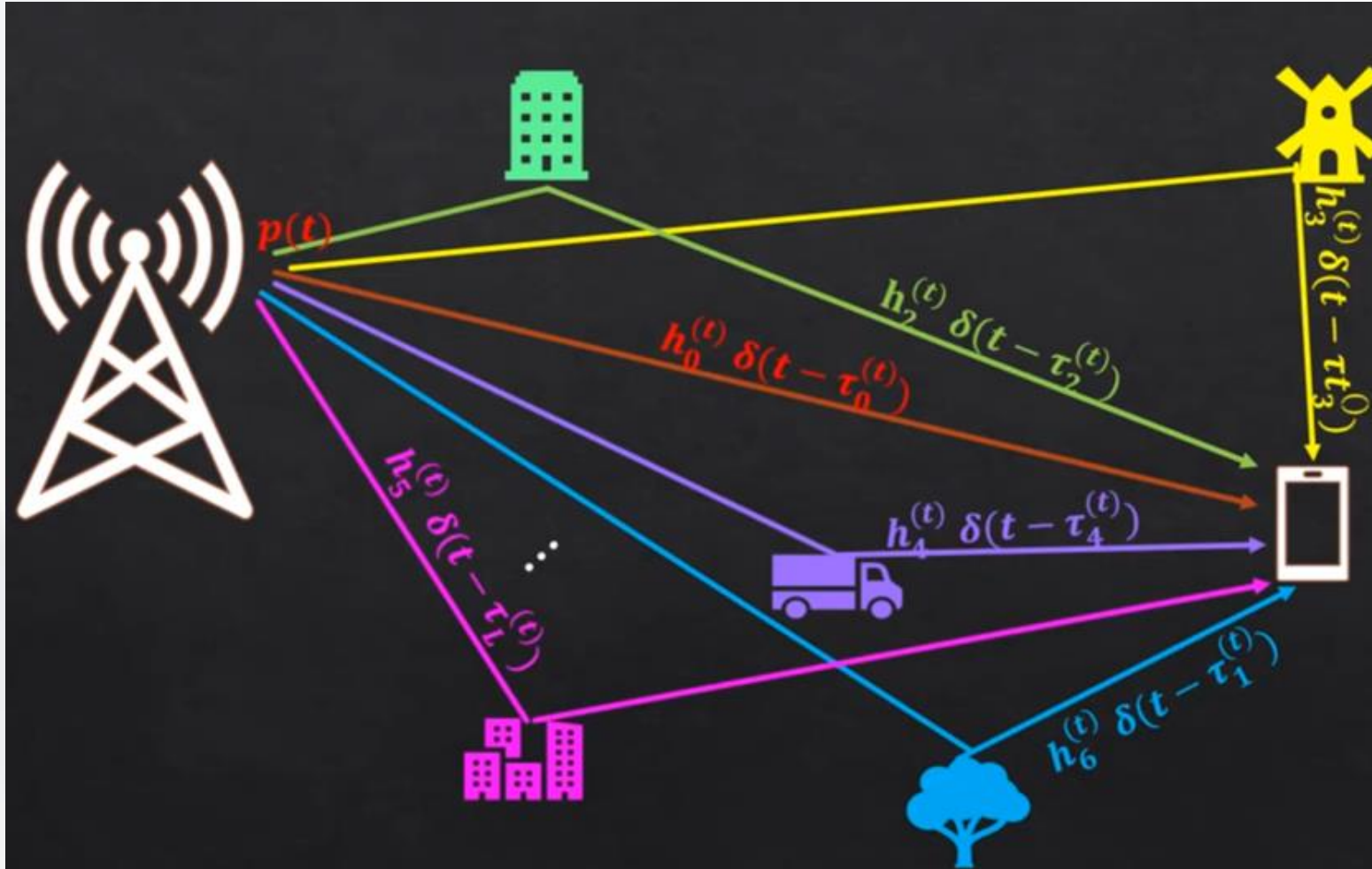
Multipath Propagation Problem

- ☑ When one signal rises while another falls, they cancel out, reducing amplitude (**destructive interference**).
- ☑ This interaction changes the shape of the resulting wave.



Multipath Propagation Problem

Time varying Multi path channel Model



- ❖ Moreover channel characteristics h_l and τ_l can change over time mainly due to mobility of MS
- ❖ Hence superscripted as $h_l^{(t)}$ and $\tau_l^{(t)}$

$$r(t) = \sum_{l=0}^L h_l^{(t)} p(t - \tau_l^{(t)})$$

Multipath Propagation Problem

☑ **Key Characteristics of Multipath Propagation:**

✓ **Multiple Signal Paths:**

- The signal reaches the receiver through direct line-of-sight (LOS) and indirect paths.

✓ **Time Delays:**

- The signals arrive at the receiver at different times.

✓ **Amplitude and Phase Variations:**

- The signals on different paths experience different attenuations and phase shifts.

✓ **Constructive and Destructive Interference:**

- The overlapping of signals can lead to constructive interference (signal enhancement) or destructive interference (signal fading), depending on the phase relationship between the signals.

Multipath Propagation Problem

☑ Effects of Multipath Propagation:

- 1) **Fading:** Rapid fluctuations in signal strength due to constructive and destructive interference.
- 2) **Intersymbol Interference (ISI):** Delayed copies of the signal interfere with subsequent symbols, causing errors in data detection.
- 3) **Reduced Signal Quality:** Degradation in SNR and overall communication performance.
- 4) **Channel Dispersion:** The channel becomes frequency-selective, affecting the performance of wideband signals.

Multipath Propagation Problem

☑ **Solutions to the Wireless Multipath Problem**

Solution Category	Techniques
Physical Layer	Diversity, Beamforming, OFDM
Signal Processing	Equalization, Rake Receiver, Channel Estimation, Error Correction Coding
System-Level	FHSS, DSSS, Adaptive Modulation and Coding
Advanced Techniques	Massive MIMO, Millimeter-Wave, Machine Learning

Multipath Propagation Problem

☑ Solutions to the Wireless Multipath Problem

Solution Category	Techniques
Physical Layer Techniques	
- Diversity Techniques	
- Spatial Diversity	Use multiple antennas (MIMO) at the transmitter or receiver.
- Frequency Diversity	Transmit the same signal over multiple frequency bands.
- Time Diversity	Transmit the same signal at different time intervals.
- Polarization Diversity	Use antennas with different polarizations to reduce correlation between paths.
- Beamforming	Use antenna arrays to focus the signal in the direction of the receiver.
- OFDM	Divide the signal into narrowband subcarriers to combat frequency-selective fading.

Multipath Propagation Problem

☑ Solutions to the Wireless Multipath Problem

Signal Processing Techniques

- Equalization	Use adaptive filters to compensate for channel distortion.
- Rake Receiver	Combine multipath components constructively in spread spectrum systems.
- Channel Estimation	Estimate and compensate for the channel impulse response.
- Error Correction Coding	Use FEC codes to detect and correct errors caused by fading.

System-Level Approaches

- Frequency Hopping (FHSS)	Rapidly switch carrier frequencies to avoid prolonged fading.
- Direct Sequence (DSSS)	Spread the signal over a wider bandwidth to reduce multipath effects.

Multipath Propagation Problem

☑ Solutions to the Wireless Multipath Problem

Advanced Techniques	
- Massive MIMO	Use a large number of antennas to exploit spatial diversity.
- Millimeter-Wave Communications	Operate at higher frequencies with reduced multipath effects.
- Machine Learning	Use AI/ML algorithms to predict and mitigate multipath effects.



Thank You